

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2022

BIOLOGY PAPER 1

8:30 am – 11:00 am (2 hours 30 minutes)

This paper must be answered in English

GENERAL INSTRUCTIONS

- (1) There are **TWO** sections, A and B, in this Paper. You are advised to finish Section A in about 35 minutes.
- (2) Section A consists of multiple-choice questions in this question paper. Section B contains conventional questions printed separately in Question-Answer Book B.
- (3) Answers to Section A should be marked on the Multiple-choice Answer Sheet while answers to Section B should be written in the spaces provided in Question-Answer Book B. **The Answer Sheet for Section A and the Question-Answer Book B for Section B will be collected separately at the end of the examination.**

INSTRUCTIONS FOR SECTION A (MULTIPLE-CHOICE QUESTIONS)

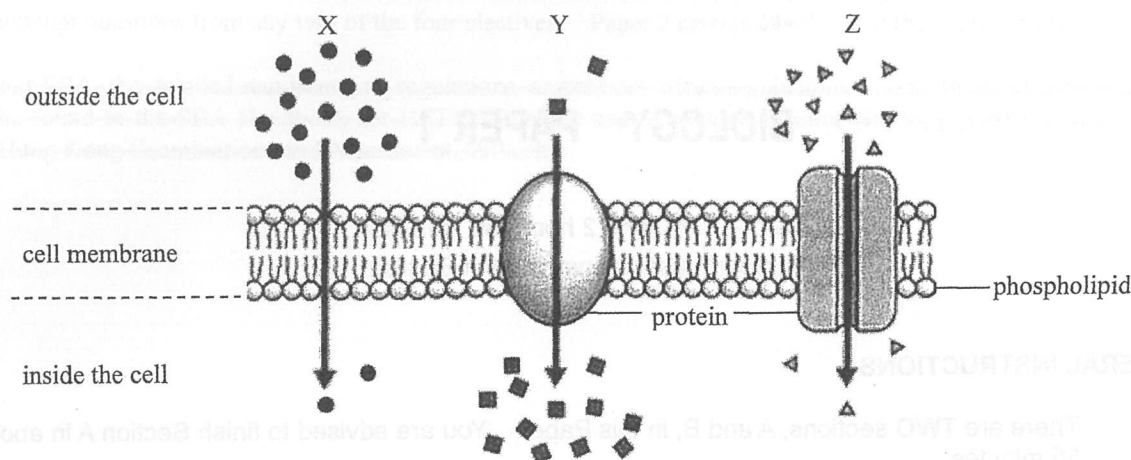
- (1) Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the 'Time is up' announcement.
- (2) When told to open this book, you should check that all the questions are there. Look for the words '**END OF SECTION A**' after the last question.
- (3) All questions carry equal marks.
- (4) **ANSWER ALL QUESTIONS.** You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.
- (5) You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.
- (6) No marks will be deducted for wrong answers.

Not to be taken away before the end of the examination session

There are 36 questions in this section.

The diagrams in this section are NOT necessarily drawn to scale.

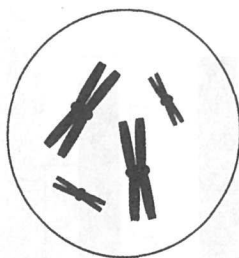
1. The diagram below shows three different ways by which substances pass through a cell membrane:



Which of the following combinations correctly matches the substances with their corresponding ways of passing through the cell membrane?

- | | | | |
|----|----------|----------|----------|
| | X | Y | Z |
| A. | oxygen | glucose | water |
| B. | glucose | water | oxygen |
| C. | water | oxygen | glucose |
| D. | water | glucose | oxygen |
2. Adenine (A) and cytosine (C) together make up 50% of the total amount of nitrogenous bases in DNA molecules. Which of the following features of DNA provide the basis to explain this phenomenon?
- (1) DNA has a helical structure.
 - (2) DNA is a double-stranded molecule.
 - (3) The nitrogenous bases are paired complementarily.
- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)
3. In Mendel's experiment, he proposed that a characteristic of the pea plants was controlled by a pair of 'heredity factors'. These factors can be separated in a 'process' before passing to the offspring. According to the current understanding of genetics, what do the 'heredity factors' and the 'process' represent respectively?
- | | | |
|----|-------------------------|------------------------------|
| | Heredity factors | Process |
| A. | alleles | first meiotic cell division |
| B. | alleles | second meiotic cell division |
| C. | chromosomes | first meiotic cell division |
| D. | chromosomes | second meiotic cell division |

Directions: Questions 4 and 5 refer to the diagram below, which shows the appearance of chromosomes in a cell at the early stage of meiotic cell division.

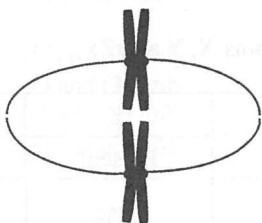


4. What is the chromosome number in the somatic cells of this organism?

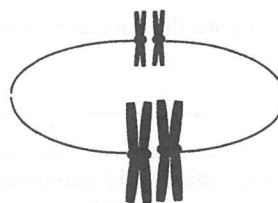
- A. 2
- B. 4
- C. 8
- D. 16

5. Which of the following diagrams correctly shows the arrangement of chromosomes found at a later stage of the cell division?

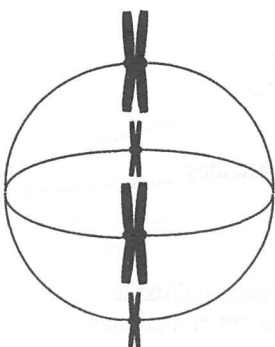
A.



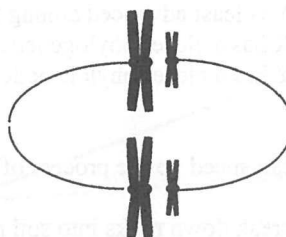
B.



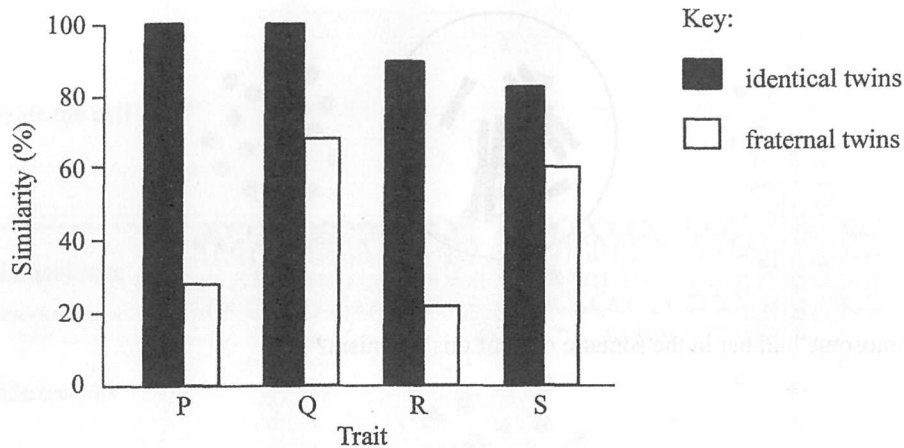
C.



D.



6. A large survey about the inheritance of four traits in identical twins and fraternal twins has been conducted. The similarity of these four traits among the individuals in each type of twins are shown in the graph below:



Which trait is most affected by environmental factors?

- A. P
- B. Q
- C. R
- D. S

7. The table below shows the comparison among three groups of organisms X, Y and Z:

	X	Y	Z
Cell wall	Present	Present	Absent
Similarity of the amino acid sequence of a protein found in all three organisms (X as a reference)	100%	45%	55%

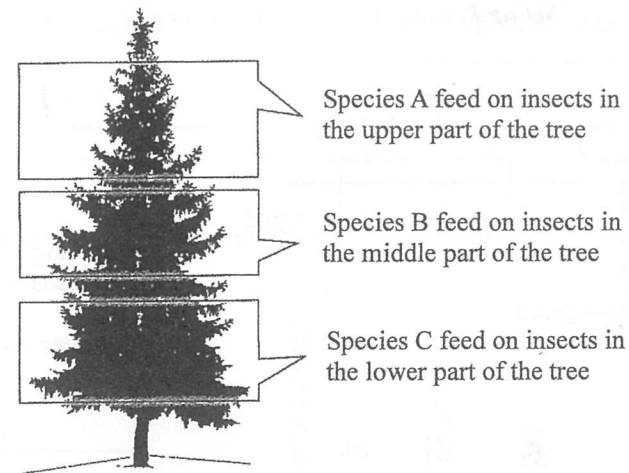
With reference to the information above, which of the following can be deduced?

- A. Both X and Y are prokaryotes.
- B. Y is least advanced among the three.
- C. X has a closer phylogenetic relationship with Z than Y.
- D. Y has a closer phylogenetic relationship with Z than X.

8. Decomposers can speed up the process of ecological succession because they

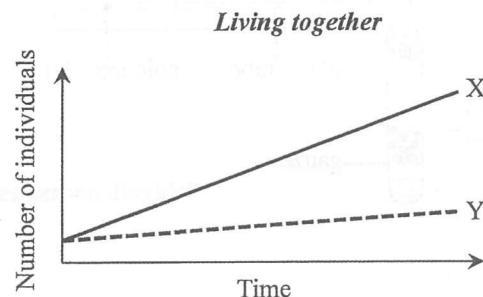
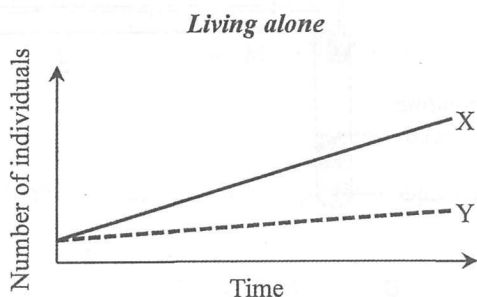
- A. break down rocks into soil particles.
- B. remove dead organic matter from the soil.
- C. release carbon back into the atmosphere in the form of carbon dioxide.
- D. release nitrogen back into the soil in the form of ammonium compounds.

9. Three bird species are living on the same tree species for years. They feed on the insects found at different regions of the tree as shown in the diagram below:



Which of the following can be deduced from the above phenomenon?

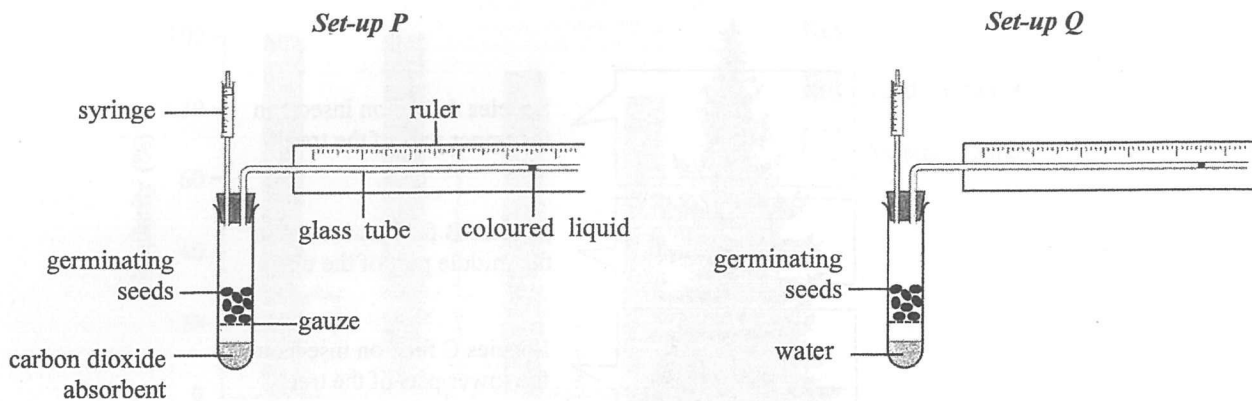
- (1) The insects are evenly distributed on the tree.
 - (2) Birds species A, B and C have different ecological niches.
 - (3) There is less interspecific competition among bird species A, B and C.
- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)
10. The graphs below show the changes in the population sizes of organisms X and Y over a period of time when they are living alone or living together.



Which of the following correctly describes the ecological relationship of X and Y?

- A. predation
 B. mutualism
 C. competition
 D. commensalism

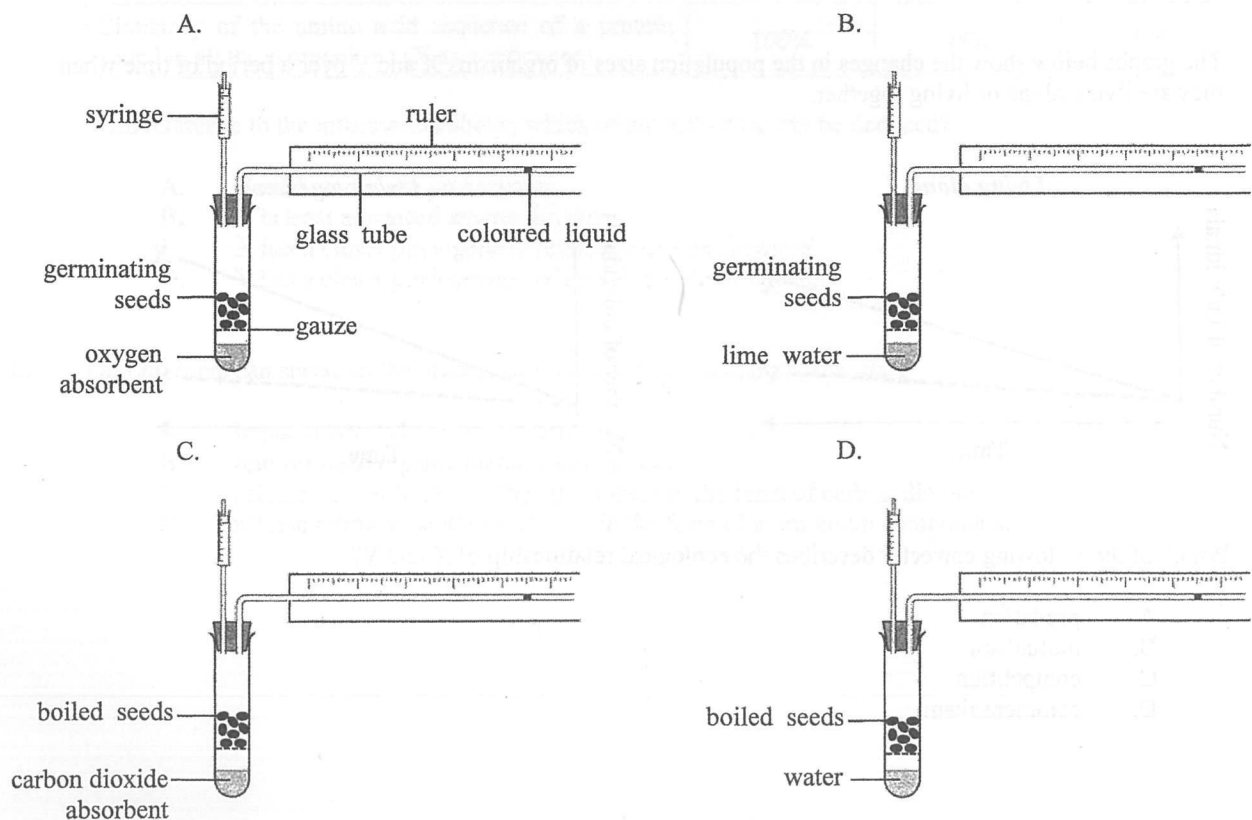
Directions: Questions 11 and 12 refer to the diagram below, which shows two set-ups used for the investigation of the gas exchange in germinating seeds:



11. Assuming that the environmental conditions are the same, what will happen to the coloured liquid in each set-up?

	<i>Set-up P</i>	<i>Set-up Q</i>
A.	moves to the right	moves to the left
B.	moves to the left	stays still
C.	stays still	moves to the left
D.	stays still	moves to the right

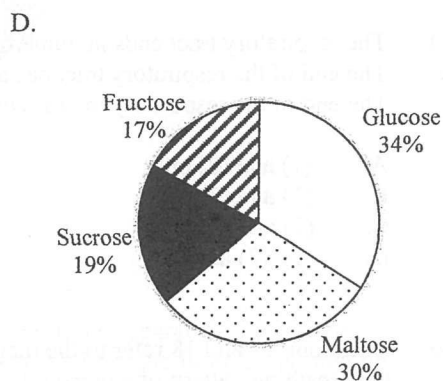
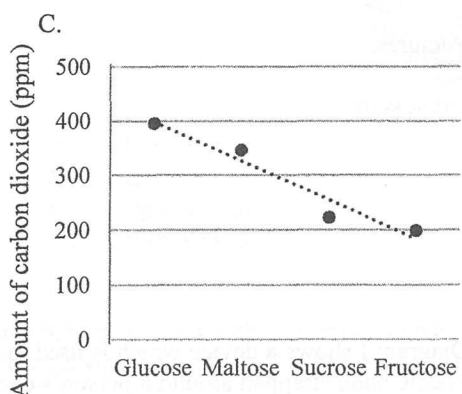
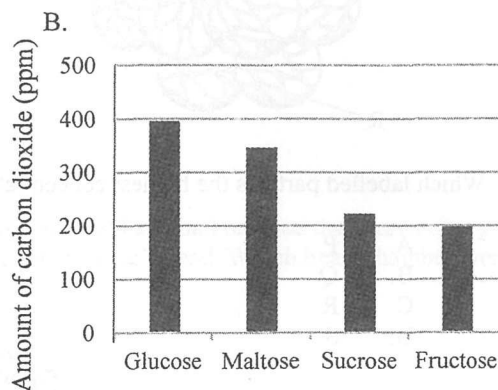
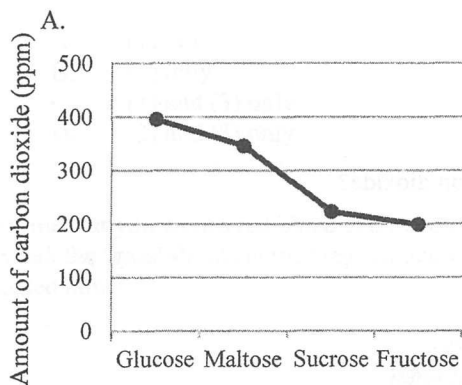
12. Which of the following modified set-ups can be used to show the identity of the gas produced in the investigation?



Directions: Questions 13 and 14 refer to the table below, which shows the effect of different substrates on the rate of anaerobic respiration of yeast:

Substrate	Average amount of carbon dioxide produced after 10 min (ppm)
Glucose	395.2
Maltose	345.8
Sucrose	222.2
Fructose	198.2

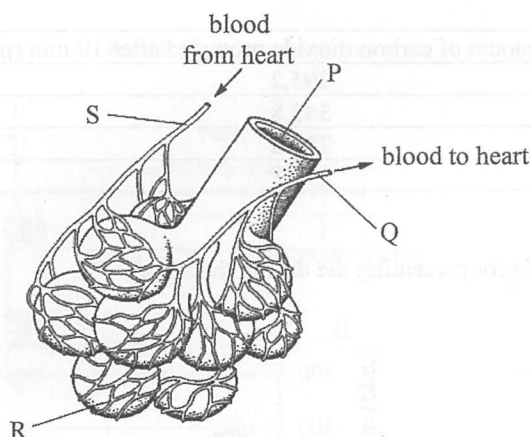
13. Which of the following graphs is most suitable for presenting the data in the table?



14. In this investigation, which part of the yeast cell produces carbon dioxide?

- A. cytoplasm
- B. cell membrane
- C. matrix of mitochondria
- D. inner membrane of mitochondria

Directions: Questions 15 and 16 refer to the diagram below, which shows the end of the respiratory tract and the associated blood vessels in humans:



15. Which labelled part has the highest concentration of carbon dioxide?

- A. P
- B. Q
- C. R
- D. S

16. Which of the following adaptive features for gas exchange can be illustrated in the above diagram?

- (1) The respiratory tract ends in numerous spherical structures.
- (2) The end of the respiratory tract has a thin wall.
- (3) The end of the respiratory tract is covered with blood vessels.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Directions: Questions 17 and 18 refer to the diagrams below. Diagram I shows a device which is used to monitor the breathing pattern of a person. It consists of an elastic band strapped around a person's chest and a sensor which detects the tension of the elastic band. Diagram II shows the change in the tension of the elastic band during breathing.

Diagram I

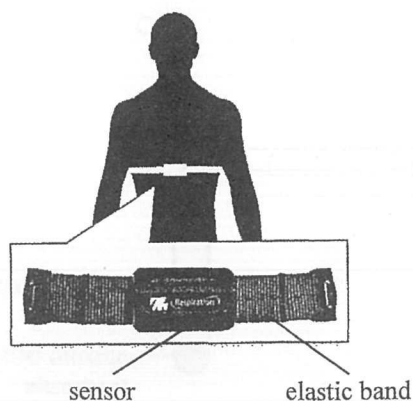
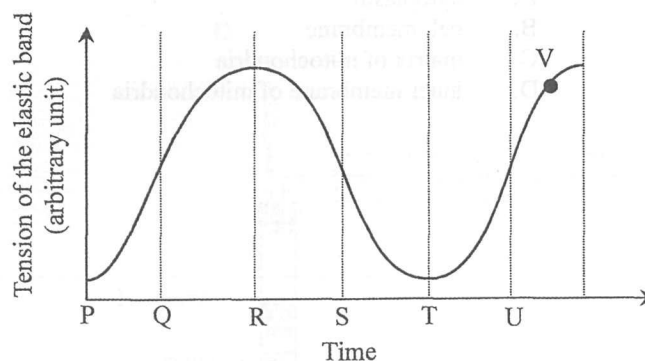
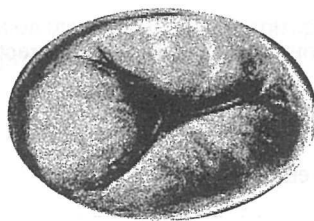


Diagram II

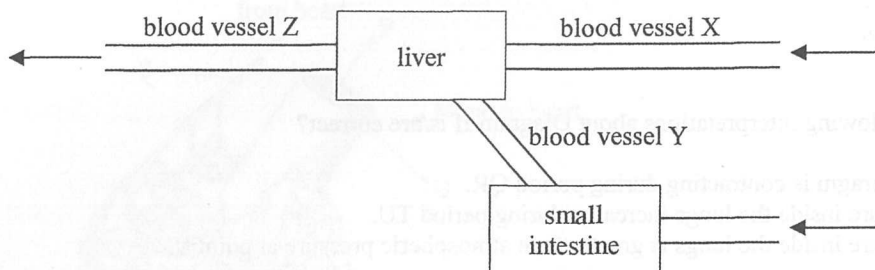


17. Exhalation takes place during period
- PR.
 - QS.
 - RT.
 - SU.
18. Which of the following interpretations about Diagram II is/are correct?
- The diaphragm is contracting during period QR.
 - Air pressure inside the lungs increases during period TU.
 - Air pressure inside the lungs is greater than atmospheric pressure at point V.
- (1) only
 - (2) only
 - (1) and (3) only
 - (2) and (3) only
19. Imagine that you were a red blood cell travelling along a blood vessel and had reached the heart. After passing through the 'gate' shown in the diagram below, there was another 'gate' ahead. Which heart chamber were you situated now?



- left atrium
 - left ventricle
 - right atrium
 - right ventricle
20. Some babies are born with a heart defect which is a hole in the septum separating the left and right ventricles. When the ventricles contract at the same time, some blood flow will deviate from the normal route. Which of the following is most likely the deviated route?
- left ventricle → right ventricle → aorta
 - left ventricle → right ventricle → pulmonary artery
 - right ventricle → left ventricle → aorta
 - right ventricle → left ventricle → pulmonary artery

Directions: Questions 21 and 22 refer to the diagram below, which shows part of the human circulatory system and the associated organs:



21. Which of the following correctly compares the blood glucose concentration in blood vessels X, Y and Z when a person is fasting?

- A. $X > Z > Y$
- B. $Y > X > Z$
- C. $Z > Y > X$
- D. $Z > X > Y$

22. Which of the following descriptions about blood vessel Y are correct?

- (1) It transports hormones.
- (2) It carries deoxygenated blood.
- (3) It has capillary networks at both ends.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

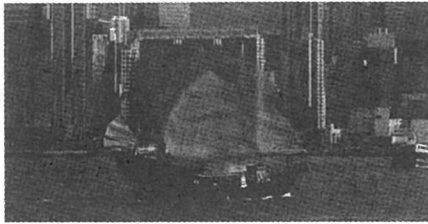
23. Which of the following structures are the components of a joint?

- (1) bone
- (2) tendon
- (3) ligament

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

24. During aging, some people may suffer from an eye defect which is caused by the degeneration of light-sensitive cells in the yellow spot. Which of the following diagrams is the most likely vision perceived by a person suffering from this eye defect?

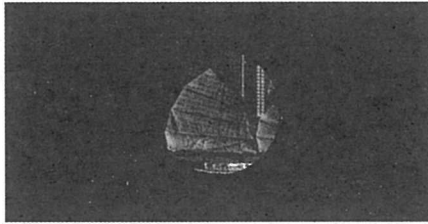
A.



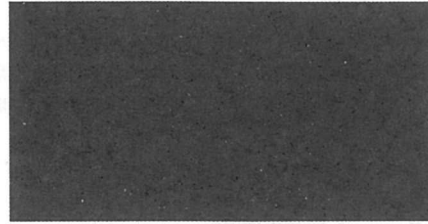
B.



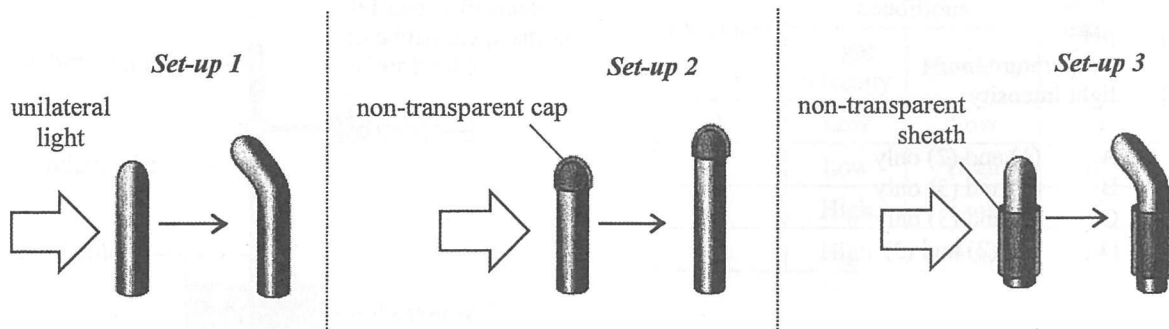
C.



D.



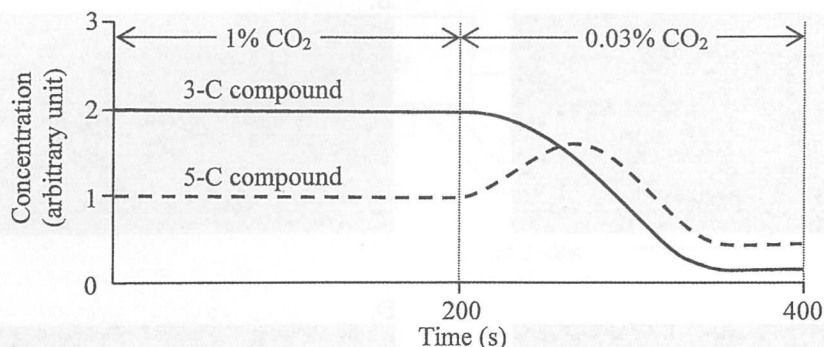
25. Darwin performed some experiments on the study of the phototropic response of coleoptiles of young seedlings. The coleoptile in each set-up was exposed to a unilateral light source and the appearance of the coleoptiles after three days are shown below:



Which of the following can be deduced from Darwin's experiment?

- (1) Auxins is produced from the tip of the coleoptile.
 - (2) The coleoptiles show positive phototropic response.
 - (3) The tip of the coleoptile detects the direction of light.
- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)

Directions: Questions 26 and 27 refer to an experiment in which a green algal culture was supplied with 1% carbon dioxide for 200 s and then followed by 0.03% carbon dioxide for another 200 s. The changes in the relative concentrations of a 3-C compound and a 5-C compound in the Calvin cycle of the green algae are shown below:



26. When the carbon dioxide concentration switched from 1% to 0.03%, which of the following combinations correctly shows the initial change of the concentration of the compound and its explanation?

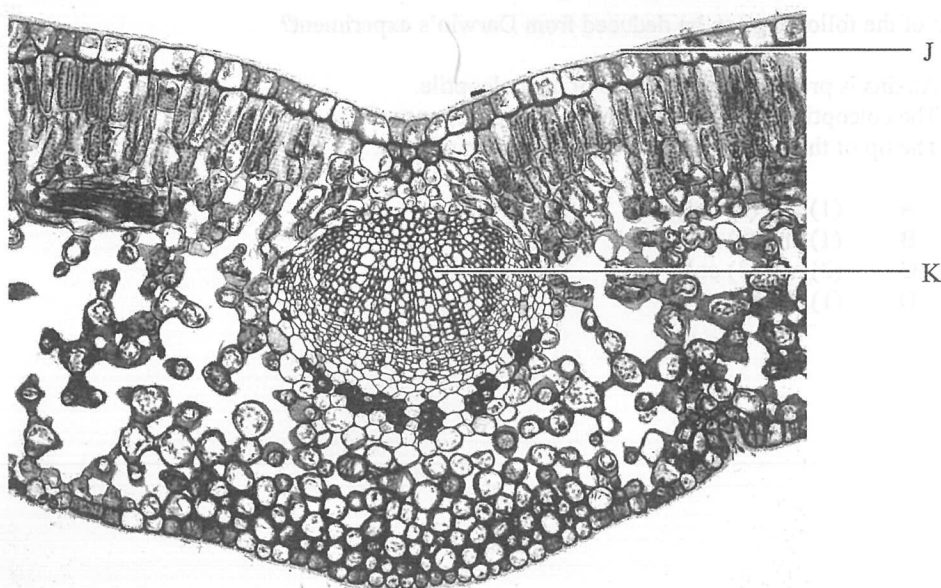
	<i>Initial change in concentration</i>	<i>Explanation</i>
A.	3-C compound decreases	reduction of 3-C compound has increased
B.	3-C compound decreases	ATP from photochemical reactions has decreased
C.	5-C compound increases	carbon fixation has decreased
D.	5-C compound increases	regeneration of carbon dioxide acceptor has increased

27. Which of the following factors should be kept constant throughout the experiment?

- (1) pH
- (2) temperature
- (3) light intensity

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Directions: Questions 28 and 29 refer to the photomicrograph below, which shows a section of a leaf:



28. Which of the following correctly describes the major function of layer J?
- It reduces water loss.
 - It allows light to pass through.
 - It protects the leaf from infection.
 - It increases the photosynthetic rate.
29. Which of the following substances will move out of the leaf through stomata from cell K in the daytime?
- water
 - oxygen
 - mineral
 - carbon dioxide

Directions: Questions 30 and 31 refer to the investigation below. Diagram I shows a set-up for measuring the transpiration rate of a leafy shoot under different environmental conditions. For each treatment, the experiment was conducted for three hours. Table II shows the initial and final readings of the position of air bubble in different treatments.

Diagram I

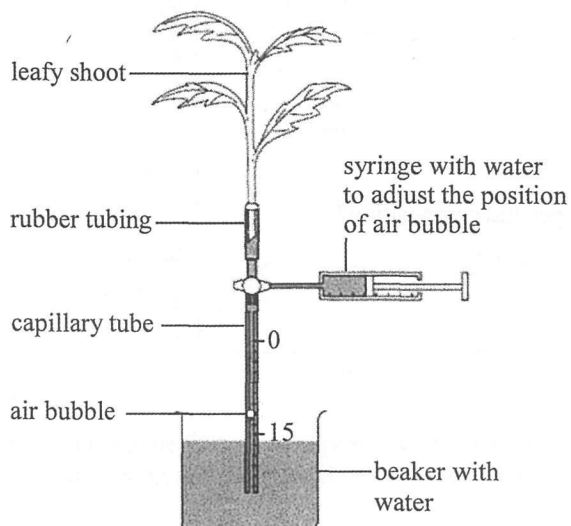
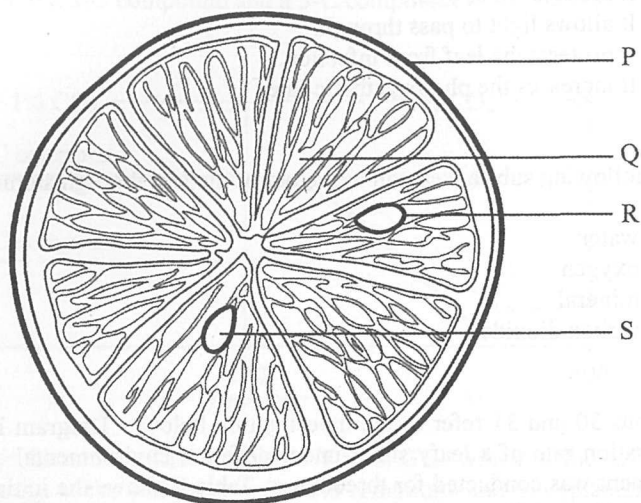


Table II

Treatment	Environmental conditions		Initial reading (cm)	Final reading (cm)
	Light intensity	Humidity		
1	Low	Low	14	7.4
2	Low	High	15	12.5
3	High	Low	15	5.6
4	High	High	14	10.6

30. In which treatment does the leafy shoot have the highest transpiration rate?
- 1
 - 2
 - 3
 - 4
31. According to the results of the investigation, which of the following changes in environmental conditions will lead to a greater reduction in the transpiration rate of the leafy shoot?
- At low humidity condition, adjust the light intensity from high to low.
 - At high humidity condition, adjust the light intensity from high to low.
 - At low light intensity, adjust the humidity from low to high.
 - At high light intensity, adjust the humidity from low to high.

Directions: Questions 32 to 34 refer to the following diagram which shows a section of a fruit:



32. Which of the following structures is developed from the ovum?

- A. P
- B. Q
- C. R
- D. S

33. Which of the following pairs of structures have the same genetic composition?

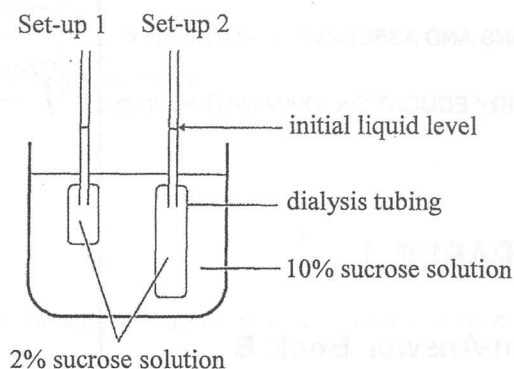
- (1) P and Q
- (2) Q and R
- (3) R and S

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

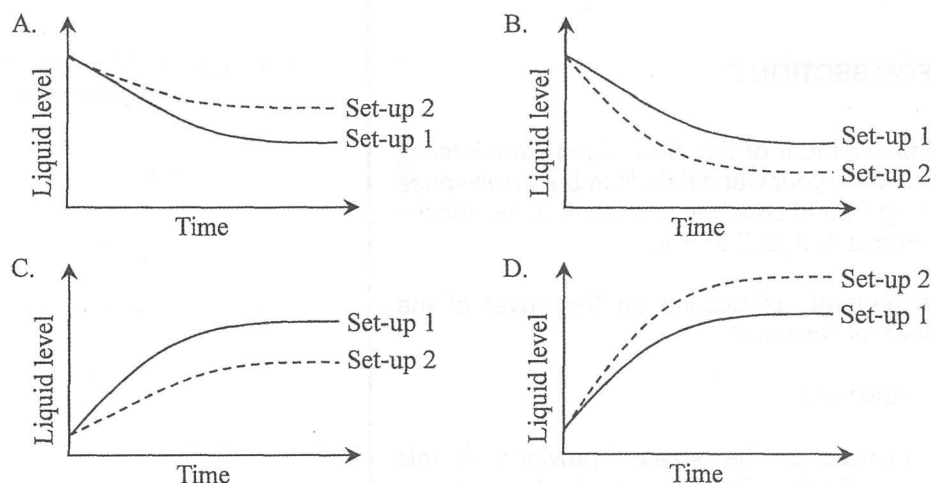
34. Which of the following statements correctly describes the major role of Q?

- A. It protects the seeds by acting as a cushion.
- B. It provides nutrients to the seeds for germination.
- C. It helps seed dispersal by attracting animals to eat the fruit.
- D. It allows the seeds to survive through adverse conditions by storing food.

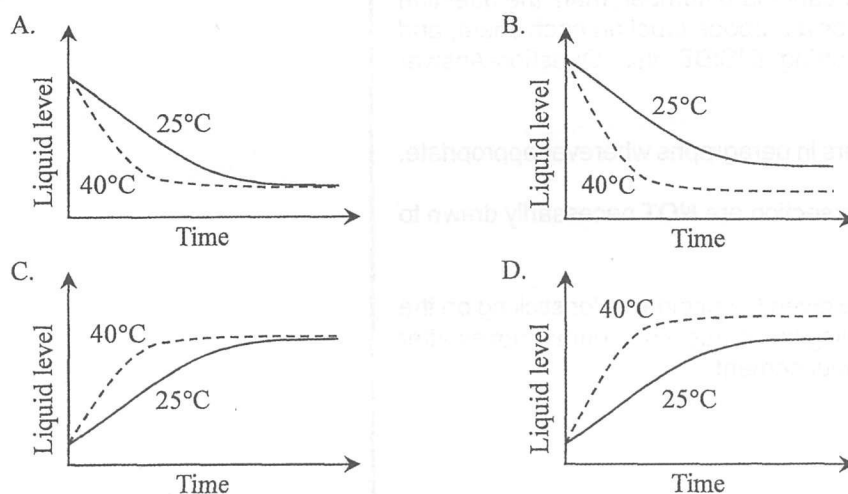
Directions: Questions 35 and 36 refer to the diagram below, which shows two set-ups used for investigating osmosis. Two dialysis tubings of different lengths were filled with 2% sucrose solution and then submerged in 10% sucrose solution. All solutions were kept at 25°C.



35. Which of the following graphs correctly shows the change in the liquid levels of set-ups 1 and 2?



36. If the experiment was repeated with the same setting except that the solutions were kept at 40°C, which of the following graphs correctly shows the change in the liquid levels of set-up 1 at different temperatures?



END OF SECTION A

Go on to Question-Answer Book B for questions on Section B